MinHeap & MaxHeap

a) Student ID: K1252 1467

$n_1 = 125$ $n_3 = 46$

$n_2 = 21$ $n_4 = 7$

Unstructured array: [107, 79, 125, 59, 21, 62, 23, 47, 46, 19, 24, 7, 6]

1) Insert 107: Index 0

2) Insert 79: Index 1. Father (107) is bigger than 79 therefore swap now 79 => Ind. 0

3) Insert 125: Index 2 the property is stored.

4) Insert 59: Index 3, Father (107). 59 < 107 swapping, new father (79). 59 < 79 swapping, 59 is the root now.

5) Insert 21: Index 4, Father (79) 21 < 79, swap, 21 < 59, swap, 21 stop 21 is at index 0 => root.

6) Insert 62: Index 5, Father (125) 62 < 125, swap, 62 > 21 occupied index 2

7) Insert 23: Index 6, Father 62, 23 < 62 swap, 23 > 21, occupied index 2

8) Insert 47: Index 7, Father (107), 47 < 107, swap 47 < 59 swap, 47 > 21, stop occupied 1

9) Insert 46: Index 8, Father (59) 46 < 59 swap, 46 < 47, swap, 46 > 21, stop occupied 1.

10) Insert 13, Index 9 Father(73) 13<73 swap, 13<46 Swap, 13<21 Swap, New root.

11) Insert 24, Index 10, Father(46) 24<46 Swap, 24>21, Stop

12) Insert 7, Index 11, Father (125), 7<125, Swap, 7<23 Swap, 7<13 swap, new root.

13) Insert 6, Index 12, Father (23), 6<23 Swap, 6<13 Swap, 6<7 swap, new root.

Index	0	1	2	3	4	5	6	7	8	9	10	11	12
107	107												
73	73	107											
125	73	107	125										
59	59	73	125	107									
21	21	59	125	107	73								
62	21	59	62	107	73	125							
23	21	59	23	107	73	125	62						
47	21	47	23	59	73	125	62	107					
46	21	46	23	47	73	125	62	107	59				
13	13	21	23	47	46	125	62	107	59	73			
24	13	21	23	47	24	125	62	107	59	73	46		
7	7	21	13	47	24	23	62	107	59	73	46	125	
6	6	21	7	47	24	13	62	107	59	73	46	125	23

b) RemoveMax()

Initial: [54, 28, 39, 8, 17, 20, 21, 5, 2, 15, 1]

Step 1: Delete 54 take 1 (index 10) and insert in the root:

[1, 28, 39, 8, 17, 20, 21, 5, 2, 15]

Step 2: Downheap the biggest child of a root is 39

$1 < 39$ swap.

Step 3: Now 1 is at Index 2 the biggest child is 21

$1 < 21$ swap.

Step 4: Now 1 is at Index 6 the children would be

$2 \cdot 6 + 1$ and $2 \cdot 6 + 2$

Index	0	1	2	3	4	5	6	7	8	9	10	Remarks
Initial	54	28	39	8	17	20	21	5	2	15	1	Initial array (MaxHeap)
Step 1	1	28	39	8	17	20	21	5	2	15		Removed max 54 Replaced root with the last
Step 2	39	28	1	8	17	20	21	5	2	15		DownHeap: Swapped 1 with max child 39
Step 3	39	28	21	8	17	20	1	5	2	15		DownHeap: Swapped 1 with max child 21.